

Consistent with this hypothesis, the transfer of information from short-term to long-term memory is enhanced by the association of new data with data previously learned and stored in long-term memory. For example, it's easier to learn a new card game if you already have “card sense” from playing other card games.

Motor skills, such as tying your shoes or writing, are usually learned by repetition. You can perform these skills without consciously recalling the individual steps required to do these tasks correctly. Learning skills and procedures, such as those required to ride a bicycle, appears to involve cellular mechanisms very similar to those responsible for brain growth and development. In such cases, neurons actually make new connections. In contrast, memorizing phone numbers, facts, and places—which can be very rapid and may require only one exposure to the relevant item—may rely mainly on changes in the strength of existing neuronal connections. Next we will consider one way that such changes in strength can take place.

Long-Term Potentiation

In searching for the physiological basis of memory, researchers have concentrated their attention on processes that can alter a synaptic connection, making the flow of communication either more efficient or less efficient. We will focus here on **long-term potentiation (LTP)**, a lasting increase in the strength of synaptic transmission. Data indicate that LTP represents a fundamental process for memory storage and learning.

First characterized in tissue slices from the hippocampus, LTP involves a presynaptic neuron that releases the excitatory neurotransmitter glutamate. LTP involves two types of glutamate receptors, each named for a molecule—NMDA or AMPA—that can be used to artificially activate that particular receptor. As shown in **Figure 49.22**, the set of receptors present on the postsynaptic membrane changes when two conditions are met: a rapid series of action potentials at the presynaptic neuron and a depolarizing stimulus elsewhere on the postsynaptic cell. The result is LTP—a stable increase in the size of the postsynaptic potentials at a synapse whose activity coincides with that of another input.

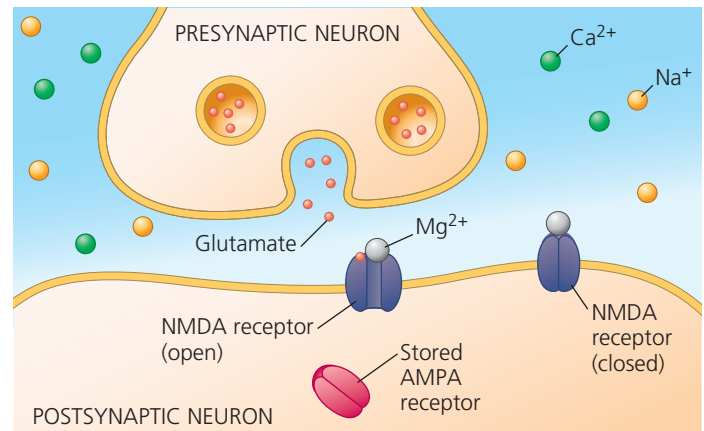
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CONCEPT CHECK 49.4

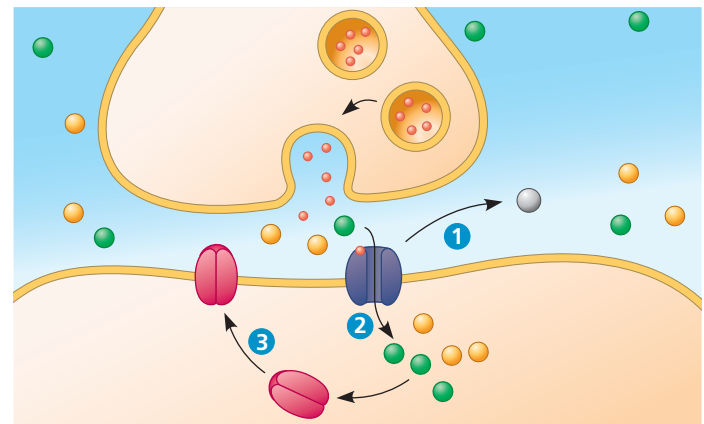
1. Outline two mechanisms by which information flow between two neurons in an adult can increase.
2. Individuals with localized brain damage have been very useful in the study of many brain functions. Why is this unlikely to be true for consciousness?
3. **WHAT IF?** Suppose that a person with damage to the hippocampus is unable to acquire new long-term memories. Why might the acquisition of short-term memories also be impaired?

For suggested answers, see Appendix A.

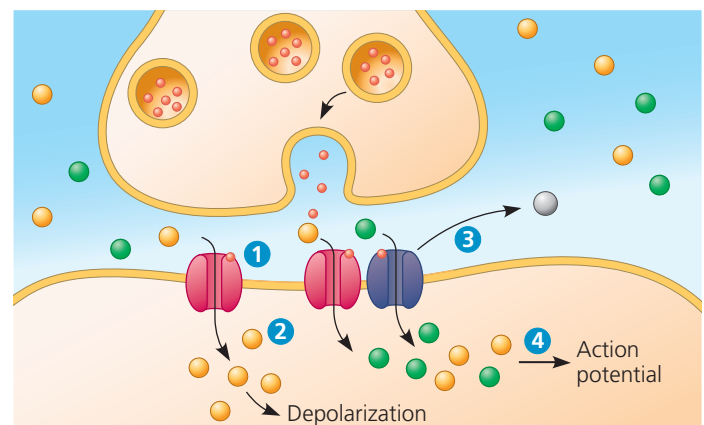
▼ **Figure 49.22 Long-term potentiation in the brain.**



(a) Synapse prior to long-term potentiation (LTP). The NMDA glutamate receptors open in response to glutamate but are blocked by Mg²⁺.



(b) Establishing LTP. Activity at nearby synapses (not shown) depolarizes the postsynaptic membrane, causing **1** Mg²⁺ release from NMDA receptors. The unblocked receptors respond to glutamate by allowing **2** an influx of Na⁺ and Ca²⁺. The Ca²⁺ influx triggers **3** insertion of stored AMPA glutamate receptors into the postsynaptic membrane.



(c) Synapse exhibiting LTP. Glutamate release activates **1** AMPA receptors that trigger **2** depolarization. The depolarization unblocks **3** NMDA receptors. Together, the AMPA and NMDA receptors trigger postsynaptic potentials strong enough to initiate **4** action potentials without input from other synapses. Additional mechanisms (not shown) contribute to LTP, including receptor modification by protein kinases.